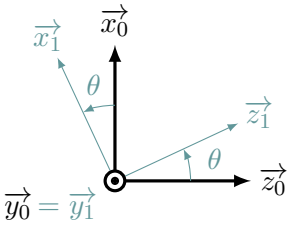
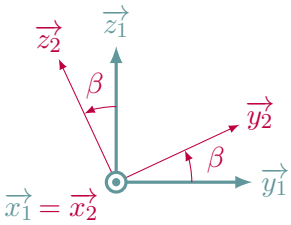
Calcul vectoriel

Exercice 1





$\overrightarrow{z_0} \cdot \overrightarrow{x_0} = 0$

$\overrightarrow{z_0} \cdot \overrightarrow{z_1} = \cos\theta$

$\overrightarrow{x_0} \cdot \overrightarrow{y_2} = \sin\theta \sin\beta$

$\overrightarrow{y_1} \cdot \overrightarrow{y_2} = \cos\beta$

$\overrightarrow{x_2} \cdot \overrightarrow{x_1} = 1$

$\overrightarrow{y_1} \cdot \overrightarrow{z_2} = -\sin\beta$

$\left[\dfrac{d\overrightarrow{z_1}}{dt}\right]_{R_0} = \dot{\theta}.\overrightarrow{x_1}$

$\left[\dfrac{d\overrightarrow{z_2}}{dt}\right]_{R_1} = -\dot{\beta}.\overrightarrow{y_2}$

$\left[\dfrac{d\overrightarrow{z_2}}{dt}\right]_{R_0} = \dot{\theta}\cos\beta.\overrightarrow{x_2} - \dot{\beta}.\overrightarrow{y_2}$

$\overrightarrow{z_0} \wedge \overrightarrow{x_0} = \overrightarrow{y_0}$

$\overrightarrow{y_0} \wedge \overrightarrow{x_1} = -\overrightarrow{z_1}$

$\overrightarrow{z_2} \wedge \overrightarrow{x_0} = \cos\beta\cos\theta.\overrightarrow{y_0} + \sin\beta.\overrightarrow{z_0}$

$\overrightarrow{z_1} \wedge \overrightarrow{y_2} = -\cos\beta\overrightarrow{x_1}$

$\overrightarrow{z_0} \wedge \overrightarrow{x_2} = \cos\theta.\overrightarrow{y_1}$

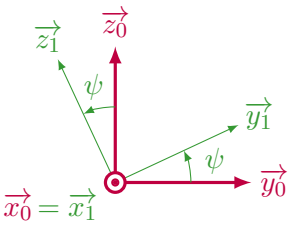
$\overrightarrow{y_2} \wedge \overrightarrow{y_1} = -\sin\beta.\overrightarrow{x_1}$

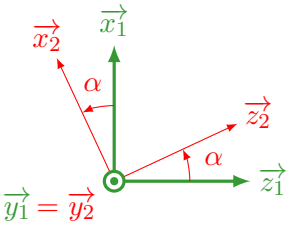
$\left[\dfrac{d\overrightarrow{x_1}}{dt}\right]_{R_0} = -\dot{\theta}.\overrightarrow{z_1}$

$\left[\dfrac{d\overrightarrow{x_0}}{dt}\right]_{R_0} = \overrightarrow{0}$

$\left[\dfrac{d\overrightarrow{y_2}}{dt}\right]_{R_0} = \dot{\theta}\sin\beta.\overrightarrow{x_1} + \dot{\beta}.\overrightarrow{z_2}$

Exercice 2





$\overrightarrow{z_0} \cdot \overrightarrow{x_0} = 0$

$\overrightarrow{z_0} \cdot \overrightarrow{z_1} = \cos\psi$

$\overrightarrow{x_0} \cdot \overrightarrow{y_2} = 0$

$\overrightarrow{y_1} \cdot \overrightarrow{y_2} = 1$

$\overrightarrow{x_2} \cdot \overrightarrow{x_1} = \cos\alpha$

$\overrightarrow{y_1} \cdot \overrightarrow{z_2} = 0$

$\left[\dfrac{d\overrightarrow{z_1}}{dt}\right]_{R_0} = -\dot{\psi}.\overrightarrow{y_1}$

$\left[\dfrac{d\overrightarrow{z_2}}{dt}\right]_{R_1} = \dot{\alpha}.\overrightarrow{x_2}$

$\left[\dfrac{d\overrightarrow{z_2}}{dt}\right]_{R_0} = -\dot{\psi}\cos\alpha.\overrightarrow{y_2} + \dot{\alpha}.\overrightarrow{x_2}$

$\overrightarrow{z_0} \wedge \overrightarrow{x_0} = \overrightarrow{y_0}$

$\overrightarrow{y_0} \wedge \overrightarrow{x_1} = -\overrightarrow{z_0}$

$\overrightarrow{z_2} \wedge \overrightarrow{x_0} = \cos\alpha.\overrightarrow{y_1}$

$\overrightarrow{x_1} \wedge \overrightarrow{y_2} = \overrightarrow{z_1}$

$\overrightarrow{z_0} \wedge \overrightarrow{x_2} = \cos\psi\cos\alpha.\overrightarrow{y_1} - \sin\psi.\overrightarrow{z_2}$

$\overrightarrow{y_2} \wedge \overrightarrow{y_1} = \overrightarrow{0}$

$\left[\dfrac{d\overrightarrow{x_1}}{dt}\right]_{R_0} = \overrightarrow{0}$

$\left[\dfrac{d\overrightarrow{y_0}}{dt}\right]_{R_1} = \dot{\psi}.\overrightarrow{z_0}$

$\left[\dfrac{d\overrightarrow{y_2}}{dt}\right]_{R_0} = \dot{\psi}.\overrightarrow{z_1}$